Quantum mechanics

Chapter IX Summary

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Planck's radiation law:

$$u(f, T) = \frac{8\pi h f^3}{c^3}\left(\frac{1}{e^{hf/k_B T} - 1}\right)$$

He could arrive at agreement with the experimental data only by making two important modifications of classical theory:

1. The oscillators (of electromagnetic origin) can only have certain discrete energies determined by

$$E_{\text{resonator}} = nhf \qquad n = 1, 2, 3, \ldots$$

where n is an integer, f is the frequency, and h is called Planck's constant and has the value

$$h = 6.6261 \times 10^{-34}\,\text{J}\cdot\text{s}$$

We want to experimentally detect the maximum value of the kinetic energy.

$$h\nu = \phi + \frac{1}{2}mv_{\max}^2$$

The retarding potentials are thus the opposing potentials needed to stop the most energetic electrons.

$$eV_0 = \frac{1}{2}mv_{\max}^2$$

The kinetic energy of the electrons depends only on the light frequency and the work function of the material.

$$eV_0 = \frac{1}{2}mv_{\max}^2 = h(\nu - \nu_0)$$

This equation can be rewritten as

$$\frac{1}{2}mv_{\max}^2 = eV_0 = h\nu - \phi$$

The frequency ν_0 represents the threshold frequency.

The expression for conservation of energy gives

$$E + m_e c^2 = E' + E_e$$

Likewise, from momentum conservation we have

$$E^2 = p^2 c^2 + m^2 c^4$$

$$p' \sin\theta = p_e \sin\phi$$

If we assume that a photon obeys the relativistic expression

$$p = p' \cos\theta + p_e \cos\phi$$

Therefore

$$E_e = hf - hf' + m_e c^2$$

$$p_e^2 = \left(\frac{hf'}{c}\right)^2 + \left(\frac{hf}{c}\right)^2 - \frac{2h^2 ff'}{c^2}\cos\theta$$

$$\lambda' - \lambda_0 = \frac{h}{m_e c}(1 - \cos\theta)$$

1924, De Broglie established the wave properties of particles. His fundamental relationship is the prediction

$$\lambda = \frac{h}{p}$$

That is, the wavelength to be associated with a particle is given by Planck's constant divided by the particle's momentum. For a photon in Einstein's special theory of relativity

$$E = pc$$

and quantum theory

$$E = h\nu$$

so

$$pc = h\nu = \frac{hc}{\lambda}$$

On partial differentiation of above equation with respect to x and t, we obtain:

$$\frac{\partial\psi(x,t)}{\partial x} = \frac{ip}{\hbar}\psi(x,t) \longrightarrow \left[\frac{\partial}{\partial x} - \frac{ip}{\hbar}\right]\psi(x,t) = 0$$

$$\frac{\partial\psi(x,t)}{\partial t} = \frac{-iE}{\hbar}\psi(x,t) \longrightarrow \left[\frac{\partial}{\partial t} + \frac{iE}{\hbar}\right]\psi(x,t) = 0$$

They suggest that

$$p = -i\hbar\frac{\partial}{\partial x} \qquad\qquad E = i\hbar\frac{\partial}{\partial t}$$

For a non-relativistic free particle, we get

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} = i\hbar\frac{\partial\psi(x,t)}{\partial t}$$

which is the Schrödinger equation for a free particle moving along the x-axis.

Probability density is defined as

$$\rho(\vec{r},t) = \Psi^*(\vec{r},t)\Psi(\vec{r},t),$$

If V(r) is real,

$$i\hbar\frac{\partial}{\partial t}\left[\Psi^*(\vec{r},t)\Psi(\vec{r},t)\right] = -\frac{\hbar^2}{2m}\left[\Psi^*(\vec{r},t)\nabla^2\Psi(\vec{r},t) - \Psi\nabla^2\Psi^*\right].$$

which yields:

$$\frac{\partial\rho(\vec{r},t)}{\partial t} + \vec{\nabla}\cdot\vec{J} = 0,$$

J(r,t) is the probability current density, or simply the current density, or even the particle density flux

$$\vec{J}(\vec{r},t) = \frac{i\hbar}{2m}\left(\Psi\vec{\nabla}\Psi^* - \Psi^*\vec{\nabla}\Psi\right)$$

The inner (or, scalar) product: The inner product (also called the scalar product) of two vectors $|\psi\rangle$ and $|\phi\rangle$ (written as $\langle\phi|\psi\rangle$) is defined by the following expression:

$$\langle\phi|\psi\rangle = \begin{pmatrix} \phi_1^* & \phi_2^* & \phi_3^* & . & . & . & \phi_n^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ . \\ . \\ . \\ \psi_n \end{pmatrix}$$

$$= (\phi_1^*\psi_1 + \phi_2^*\psi_2 + \phi_3^*\psi_3 + ... + \phi_n^*\psi_n) = \sum_{i=1}^{n} \phi_i^*\psi_i.$$

We call $\langle\phi|\psi\rangle$ a 'bracket'.

$$\langle\phi \mid \psi\rangle = \int \phi^*(\vec{r}, t)\,\psi(\vec{r}, t)\, d^3r.$$

The action of an arbitrary operator A on a ket $|\psi\rangle \in V$ is written as:

$$\hat{A}\,|\psi\rangle = |\phi\rangle\,.$$

The product of an operator A and a number (complex) a is an operator aA, which takes a vector $|\psi\rangle \in V$ into the vector $a(A\psi\rangle) \in V$:

$$(a\hat{A})|\psi\rangle = a(\hat{A}|\psi\rangle.$$

The sum, C, of two operators A and B is defined as

$$\hat{C}\,|\psi\rangle = (\hat{A}+\hat{B})\,|\psi\rangle = \hat{A}\,|\psi\rangle + \hat{B}\,|\psi\rangle\,.$$

The operators in quantum mechanics are linear.

$$\hat{A}(\alpha\,|\psi\rangle + \beta\,|\phi\rangle) = \alpha\hat{A}\,|\psi\rangle + \beta\hat{A}\,|\phi\rangle\,.$$

The Hermitian adjoint, or simply the adjoint, $A^\dagger$, of an operator A is defined by this relation:

$$\langle \psi \mid \hat{A}^\dagger \mid \phi \rangle = \langle \phi \mid \hat{A} \mid \psi \rangle^*.$$

An operator A is said to be *Hermitian* if it is equal to its adjoint $A^\dagger$:

$$\boxed{\hat{A} = \hat{A}^\dagger \qquad \text{or} \quad \langle \psi \mid \hat{A} \mid \phi \rangle = \langle \phi \mid \hat{A} \mid \psi \rangle^*.}$$

On the other hand, an operator B is said to be *skew-Hermitian* or *anti-Hermitian* if

$$\boxed{\hat{B}^\dagger = -\hat{B} \qquad \text{or} \quad \langle \psi \mid \hat{B} \mid \phi \rangle = -\langle \phi \mid \hat{B} \mid \psi \rangle^*.}$$

The *commutator* of two operators A and B, denoted by $[A, B]$, is defined by

$$[\hat{A},\ \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A},$$

and the anticommutator {A, B} is defined by

$$\{\hat{A},\ \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.$$

We can establish the following properties.

Antisymmetry: $$[\hat{A},\ \hat{B}] = -[\hat{B},\ \hat{A}]$$

Linearity: $$[\hat{A},\ \hat{B} + \hat{C} + \hat{D} + \cdots] = [\hat{A},\ \hat{B}] + [\hat{A},\ \hat{C}] + [\hat{A},\ \hat{D}] + \cdots$$

Hermitian conjugate of a commutator: $$[\hat{A},\ \hat{B}]^\dagger = [\hat{B}^\dagger,\ \hat{A}^\dagger]$$

Distributivity: $$[\hat{A},\ \hat{B}\hat{C}] = [\hat{A},\ \hat{B}]\hat{C} + \hat{B}[\hat{A},\ \hat{C}]$$

$$[\hat{A}\hat{B},\ \hat{C}] = \hat{A}[\hat{B},\ \hat{C}] + [\hat{A},\ \hat{C}]\hat{B}$$

Since the last term is a positive real number, we can infer the following relation:

$$\left|\langle \Delta\hat{A}\Delta\hat{B}\rangle\right|^2 \geq \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2. \qquad \Delta A = \sqrt{\langle(\Delta\hat{A})^2\rangle} = \sqrt{\langle\hat{A}^2\rangle - \langle\hat{A}\rangle^2},$$

and

$$\langle(\Delta\hat{A})^2\rangle\langle(\Delta\hat{B})^2\rangle \geq \frac{1}{4}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|^2,$$

Finally

$$\boxed{\Delta A\Delta B \geq \frac{1}{2}\left|\langle[\hat{A},\ \hat{B}]\rangle\right|.}$$

This uncertainty relation plays an important role in the formalism of quantum mechanics for position and momentum operators

$$\boxed{\Delta x\Delta p_x \geq \frac{\hbar}{2}, \qquad \Delta y\Delta p_y \geq \frac{\hbar}{2}, \qquad \Delta z\Delta p_z \geq \frac{\hbar}{2}.}$$

Since

$$\boxed{[\hat{X},\ \hat{P}_x] = i\hbar\hat{I}, \qquad [\hat{Y},\ \hat{P}_y] = i\hbar\hat{I}, \qquad [\hat{Z},\ \hat{P}_z] = i\hbar\hat{I},}$$

A state vector $|\psi\rangle$ is said to be an *eigenvector* (also called an eigenket or eigenstate) of an operator A if the application of A to $|\psi\rangle$ gives

$$\hat{A} \mid \psi\rangle = a \mid \psi\rangle,$$

For a Hermitian operator, all of its eigenvalues are real and the eigenvectors corresponding to different eigenvalues are orthogonal.

If $\hat{A}^\dagger = \hat{A}$, $\quad \hat{A} \mid \phi_n\rangle = a_n \mid \phi_n\rangle \quad \Longrightarrow \quad a_n =$ real number, and $\langle \phi_m \mid \phi_n\rangle = \delta_{mn}$.

Since $\quad \hat{A} \mid \phi_n\rangle = a_n \mid \phi_n\rangle \quad \Longrightarrow \quad \langle \phi_m \mid \hat{A} \mid \phi_n\rangle = a_n \langle \phi_m \mid \phi_n\rangle,$

$$\langle \phi_m \mid \hat{A}^\dagger = a_m^* \langle \phi_m \mid \quad \Longrightarrow \quad \langle \phi_m \mid \hat{A}^\dagger \mid \phi_n\rangle = a_m^* \langle \phi_m \mid \phi_n\rangle.$$

$$(a_n - a_m^*)\langle \phi_m \mid \phi_n\rangle = 0.$$

For each linear operator A, we can write

$$\hat{A} = \hat{I}\hat{A}\hat{I} = \left(\sum_{n=1}^{\infty} |\phi_n\rangle\langle\phi_n|\right)\hat{A}\left(\sum_{m=1}^{\infty} |\phi_m\rangle\langle\phi_m|\right) = \sum_{nm} A_{nm} |\phi_n\rangle\langle\phi_m|,$$

where A_{nm} is the nm matrix element of the operator A:

$$A_{nm} = \langle\phi_n | \hat{A} | \phi_m\rangle.$$

We see that the operator A is represented, within the basis $\{|\phi_n\rangle\}$, by a *square* matrix A

$$A = \begin{pmatrix} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ A_{31} & A_{32} & A_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$

In the position representation, the basis consists of an infinite set of vectors $|r\rangle$ which are eigenkets to the position operator R:

$$\hat{\vec{R}} \mid \vec{r}\rangle = \vec{r} \mid \vec{r}\rangle,$$

So every state vector can be expanded as follows:

$$\mid \psi\rangle = \int d^3 r \mid \vec{r}\rangle\langle\vec{r} \mid \psi\rangle \equiv \int d^3 r \, \psi(\vec{r}) \mid \vec{r}\rangle,$$

where

$$\langle\vec{r} \mid \psi\rangle = \psi(\vec{r}).$$

This is known as the *wave function* for the state vector. The scalar product between two state vectors, can be expressed in this form:

$$\langle\phi \mid \psi\rangle = \langle\phi \mid \left(\int d^3 r \mid \vec{r}\rangle\langle\vec{r} \mid\right) \mid \psi\rangle = \int d^3 r \, \phi^*(\vec{r})\psi(\vec{r}).$$

The basis $|p\rangle$ of the momentum representation is obtained from the eigenkets of the momentum operator P:

$$\hat{\vec{P}} \mid \vec{p}\rangle = \vec{p} \mid \vec{p}\rangle,$$

Expanding the vector state in this basis, we obtain

$$\mid \psi\rangle = \int d^3p \mid \vec{p}\rangle\langle\vec{p} \mid \psi\rangle = \int d^3p\, \Psi(\vec{p}) \mid \vec{p}\rangle,$$

where the expansion coefficient $\Psi(p)$ represents the *momentum space wave function*.

$$\boxed{\langle\vec{p} \mid \vec{r}\rangle = \langle\vec{r} \mid \vec{p}\rangle^* = \frac{1}{(2\pi\hbar)^{3/2}} e^{-i\vec{p}\cdot\vec{r}/\hbar}.}$$

We can write

$$\Psi(\vec{p}) = \langle\vec{p} \mid \psi\rangle = \langle\vec{p} \mid \int d^3r \mid \vec{r}\rangle\langle\vec{r} \mid \psi\rangle = \int d^3r \langle\vec{p} \mid \vec{r}\rangle \psi(\vec{r}).$$

Average value of a dynamical variable: The average value, $\langle A \rangle$, of a dynamical variable A, in a given state ψ of the system, is defined as

$$\langle A \rangle = \int_{-\infty}^{+\infty} \psi^*(\vec{r}) \left[\hat{A}\psi(\vec{r})\right] d^3x \Big/ \int_{-\infty}^{+\infty} \psi^*(\vec{r})\psi(\vec{r})\, d^3x,$$

where the integration is over the entire region of variation of the independent variables, x, y, and z. The asterisk stands for complex conjugation.

If the wave function is normalized to unity, the required average value is given by

$$\langle A \rangle = \int_{-\infty}^{+\infty} \psi^*(\vec{r})(\hat{A}\psi(\vec{r}))\, d^3x.$$

The propagator can be determined as follows,

$$i\hbar\frac{\partial\hat{U}(t,t_0)}{\partial t}=\hat{H}\hat{U}(t,t_0).$$

If the Hamiltonian, H, is time independent, its solution satisfying the initial condition, $U(t_0,t_0) = I$, can be written as

$$\hat{U}(t,t_0)=e^{-\frac{i}{\hbar}(t-t_0)\hat{H}}.$$

The wave function can be written as

$$|\psi(t)\rangle=e^{-\frac{i}{\hbar}(t-t_0)\hat{H}}|\psi(t_0)\rangle.$$

We can expand the wave function $\psi(q,0)$ into a series with respect to the eigenfunctions, $\phi_m(q)$, m = 1,2,3,..., of the Hamiltonian

$$\psi(q,t)=\sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{-i\hat{H}}{\hbar}(t-t_0)\right)^n\sum_m c_m\phi_m$$

$$\psi(q,t_0)=\sum_m c_m\phi_m(q),$$

$$=\sum_m c_m\phi_m\sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{-iE_m^0}{\hbar}(t-t_0)\right)^n=\sum_m c_m\phi_m e^{-\frac{i}{\hbar}E_m^0(t-t_0)}.$$

It can be written as

$$\frac{d\langle\hat{A}\rangle}{dt} = \frac{\partial\langle A\rangle}{\partial t} + \frac{1}{i\hbar}\langle[\hat{A},\hat{H}]\rangle,$$

In the case when there is no explicit dependence of the operator A on time, we have

$$\frac{d\langle\hat{A}\rangle}{dt} = \frac{1}{i\hbar}\langle[\hat{A},\hat{H}]\rangle.$$

Ehrenfest's theorem: The average values of observables in quantum mechanics obey the classical equations of motion.

$$\frac{d\langle\hat{x}\rangle}{dt} = \frac{\langle\hat{p}_x\rangle}{m}. \qquad \frac{d\langle\hat{p}_x\rangle}{dt} = -\left\langle\frac{\partial V(x)}{\partial x}\right\rangle.$$

The general stationary state solution is the linear superposition given by

$$\psi(x,t) = A_{(+)}e^{i(kx-\omega t)} + A_{(-)}e^{-i(kx+\omega t)},$$

where $A_{(+)}$ and $A_{(-)}$ are arbitrary, in general complex, constants. If we use the de Broglie formula, the solution can be written as

$$\psi(x,t) = A_{(+)}\, e^{\frac{i}{\hbar}(px-Et)} + A_{(-)}\, e^{-\frac{i}{\hbar}(px+Et)}.$$

The first term in the above equation represents a particle traveling to the right (positive x direction) and the second term represents a particle traveling to the left.

The potential is given by the following expression:

$$V(x) = \begin{cases} 0, & \text{for } 0 < x < a, \\ \infty, & \text{for } x \leq 0, x \geq a. \end{cases}$$

The normalized eigenfunctions and the corresponding energies of infinite potential well are

$$\psi_n(x,t) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a} n\right), \; E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}, \; n = 1, 2, 3, \ldots .$$

The full stationary state solutions are

$$\psi_n(x,t) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a} n\right) e^{-i\frac{n^2\pi^2\hbar}{2ma^2}t}.$$

If the potential is finite jump

$$\lim_{\varepsilon\to 0}\int_{-\varepsilon}^{+\varepsilon} V(x)\phi(x)\,dx. = V_0 \lim_{\varepsilon\to 0}\int_{-\varepsilon}^{+\varepsilon} \phi(x)\,dx = 0.$$

If the potential is delta potential

$$\left(\frac{d\phi}{dx}\right)_{+\varepsilon} - \left(\frac{d\phi}{dx}\right)_{-\varepsilon} = -\frac{2m\alpha}{\hbar^2}\int_{-\varepsilon}^{+\varepsilon}\delta(x-x_0)\phi(x)\,dx = -\frac{2m\alpha}{\hbar^2}\phi(x_0).$$

If the eigenvalue and wavefuction of delta function is

$$E = -\frac{m\alpha^2}{2\hbar^2} \qquad \phi(x) = \begin{cases} \sqrt{k}e^{kx} & \text{for} \quad x<0, \\ \sqrt{k}e^{-kx} & \text{for} \quad x>0. \end{cases} \qquad \text{or,} \qquad \phi(x) = \sqrt{\frac{m\alpha}{\hbar^2}}\,e^{-\frac{m\alpha}{\hbar^2}|x|}.$$

Let ξ_n be the n^{th} root of the transcendental equations. If we introduce the notation

$$\xi_n^2 = (k_1 a)^2 = \frac{2ma^2 E_n}{\hbar^2},$$

then

$$\eta = \sqrt{R^2 - \xi_n^2}$$

and the transcendental equations take the form

$$\xi_n \tan \xi_n = \sqrt{R^2 - \xi_n^2}. \qquad \text{(For even parity states)}$$

$$-\xi_n \cot \xi_n = \sqrt{R^2 - \xi_n^2}. \qquad \text{(For odd parity states)}$$

Consider the one-dimensional simple harmonic oscillator characterized by the potential energy

$$V(x) = \frac{1}{2} m\omega^2 x^2,$$

The quantization of energy of the harmonic oscillator is:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, 3, \ldots$$

The wave function is

$$\phi_n(\xi) = N_n e^{-\xi^2/2} H_n(\xi), \qquad \xi = \sqrt{\alpha} x, \qquad \alpha = \frac{m\omega}{\hbar}.$$

and

$$N_n = \sqrt{\frac{\alpha^{1/2}}{2^n n! \pi^{1/2}}}.$$

The reflection coefficient is defined as

$$\mathscr{R} = \frac{\text{Reflected particle flux density}}{\text{Incident particle flux density}} = \frac{J_R}{J_I} = \frac{v_1|B|^2}{v_1|A|^2} = \frac{|B|^2}{|A|^2}.$$

It is given by

$$\mathscr{R} = \frac{\left(\frac{k_2^2+k_1^2}{k_2k_1}\right)^2 \sinh^2(k_2a)}{\left[4\cosh^2(k_2a) + \left(\frac{k_2^2-k_1^2}{k_1k_2}\right)^2 \sinh^2(k_2a)\right]}.$$

The transmission coefficient, on the other hand, is defined as

$$\mathscr{T} = \frac{\text{Transmitted particle flux density}}{\text{Incident particle flux density}} = \frac{J_T}{J_I} = \frac{|F|^2}{|A|^2}.$$

and

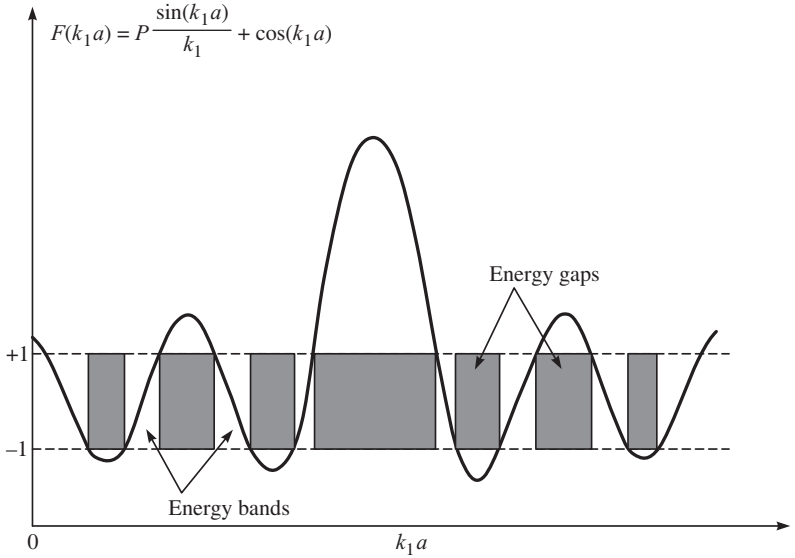
$$\mathscr{T} = \frac{4}{\left[4\cosh^2(k_2a) + \left(\frac{k_2^2-k_1^2}{k_1k_2}\right)^2 \sinh^2(k_2a)\right]}.$$

The Bloch's theorem

$$\psi(x+c) = e^{i\kappa(x+c)} u_\kappa(x+c) = e^{i\kappa c}\, e^{i\kappa x}\, u_\kappa(x) = e^{i\kappa c}\, \psi(x).$$

It yields the following transcendental equation for the determination of the energy eigenvalues

$$\frac{(k_2^2 - k_1^2)}{2k_1 k_2} \sinh(k_2 b)\, \sin(k_1 a) + \cosh(k_2 b)\, \cos(k_1 a) = \cos[K(a+b)].$$



The quantum mechanical operator for L, is obtained by replacing r and p, with their respective operators,

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}.$$

The components of angular momentum operator can be expressed in Cartesian Coordinates

$$\hat{L}_x = y\,\hat{p}_z - z\,\hat{p}_y = -i\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right),$$

$$\hat{L}_y = z\,\hat{p}_x - x\,\hat{p}_z = -i\hbar\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}\right),$$

$$\hat{L}_z = x\,\hat{p}_y - y\,\hat{p}_x = -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

Therefore,

$$[\hat{L}_x, \hat{L}_y] = i\hbar\,(x\,\hat{p}_y - y\,\hat{p}_x) = i\hbar\hat{L}_z.$$

The other two commutators are calculated in a similar manner. The net result is

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \qquad [\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x, \qquad [\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y.$$

These commutation relations can be combined together into a single vector equation

$$i\hbar\hat{\vec{L}} = \hat{\vec{L}} \times \hat{\vec{L}}.$$

Equivalently, they can also be written as

$$\left[\hat{L}_j, \hat{L}_k\right] = i\hbar\,\varepsilon_{jk\ell}\,\hat{L}_\ell,$$

Let us introduce the operators:

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y.$$

Using the commutation relations

$$[\hat{L}_z, \hat{L}_+] = [\hat{L}_z, \hat{L}_x] + i[\hat{L}_z, \hat{L}_y] = i\hbar\hat{L}_y + i(-i)\hbar\hat{L}_x = \hbar(\hat{L}_x + i\hat{L}_y) = \hbar\hat{L}_+,$$

$$[\hat{L}_z, \hat{L}_-] = [\hat{L}_z, \hat{L}_x] - i[\hat{L}_z, \hat{L}_y] = i\hbar\hat{L}_y - \hbar\hat{L}_x = -\hbar(\hat{L}_x - i\hat{L}_y) = -\hbar\hat{L}_-.$$

Hence,

$$[\hat{L}^2, \hat{L}_\pm] = [\hat{L}_x^2, \hat{L}_\pm] + [\hat{L}_y^2, \hat{L}_\pm] + [\hat{L}_z^2, \hat{L}_\pm] = [\hat{L}_x^2, \hat{L}_x \pm i\hat{L}_y]$$

$$+ [\hat{L}_y^2, \hat{L}_x \pm i\hat{L}_y] + [\hat{L}_z^2, \hat{L}_x \pm i\hat{L}_y] = \pm i[\hat{L}_x^2, \hat{L}_y] + [\hat{L}_y^2, \hat{L}_x]$$

$$+ [\hat{L}_z^2, \hat{L}_x] \pm i[\hat{L}_z^2, \hat{L}_y] = \mp\hbar\left(\hat{L}_x\hat{L}_z + \hat{L}_z\hat{L}_x\right) - i\hbar\left(\hat{L}_y\hat{L}_z + \hat{L}_z\hat{L}_y\right)$$

$$+ i\hbar\left(\hat{L}_y\hat{L}_z + \hat{L}_z\hat{L}_y\right) \pm \hbar\left(\hat{L}_x\hat{L}_z + \hat{L}_z\hat{L}_x\right) = 0.$$

The eigenvalue equations for L^2 and L_z, respectively, are

$$\hat{L}^2 Y_{lm}(\theta,\varphi) = \hbar^2 l(l+1) Y_{lm}(\theta,\varphi),$$

$$\hat{L}_z Y_{lm}(\theta,\varphi) = m\hbar Y_{lm}(\theta,\varphi).$$

where l = 0, 1, 2, 3,... and m = -l,−l+1,−l+2,−l+3,...,0,1,2,3,...,l−1,l.

The full normalized eigenfunctions of L^2 and L_z are now given by

$$Y_{lm}(\theta,\varphi) = (-1)^m \sqrt{\left(\frac{2l+1}{4\pi}\right)\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\theta) e^{im\varphi} \qquad (m \geq 0).$$

and

$$Y_{lm}(\theta,\varphi) = \sqrt{\left(\frac{2l+1}{4\pi}\right)\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\theta) e^{im\varphi} \qquad m < 0.$$

$$\int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \, \sin\theta \, Y_{\ell'}^{m'} \hat{L}^2 Y_\ell^m = \hbar^2 \ell(\ell+1)\delta_{\ell'\ell}\delta_{m'm},$$

$$\int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \, \sin\theta \, Y_{\ell'}^{m'} \hat{L}_z Y_\ell^m = m\hbar\delta_{\ell'\ell}\delta_{m'm}.$$

The matrices, corresponding to L_+ and L_- in this basis, are calculated to be

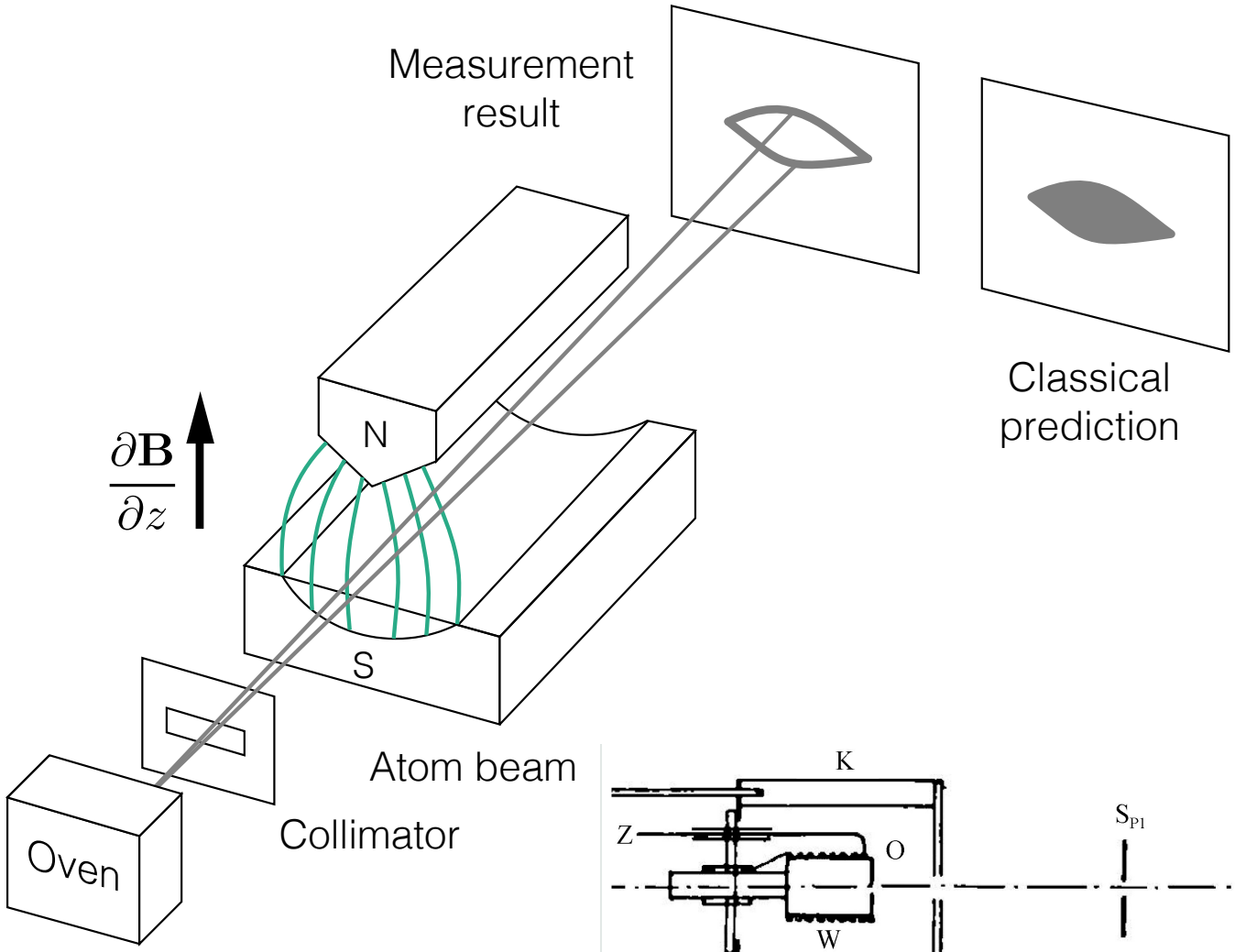
$$L_+ = \sqrt{2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad L_- = \sqrt{2}\hbar \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

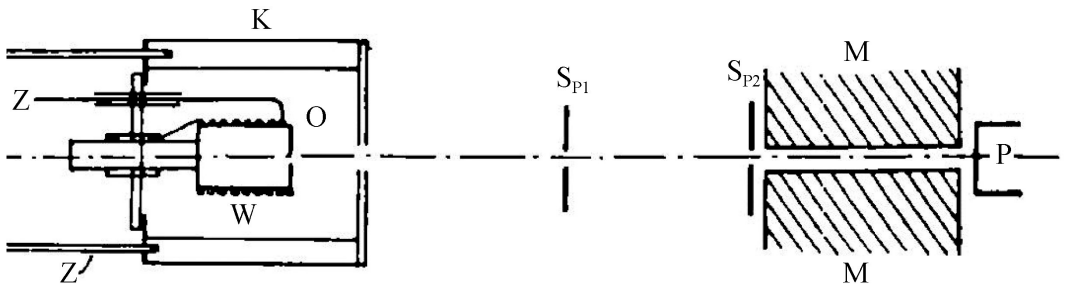
Taking into account that

$$\hat{L}_x = (\hat{L}_+ + \hat{L}_-)/2 \text{ and } \hat{L}_y = (\hat{L}_+ - \hat{L}_-)/2i,$$

we get

$$L_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$


Measurement result
Classical prediction
$\frac{\partial \mathbf{B}}{\partial z}$
N
S
Atom beam
Collimator
Oven


K
Z
O
W
Z'
S_{P1}
S_{P2}
M
P
M

Spin is denoted by a vector S. As required by the rules of quantum mechanics, it is represented by an operator S with Cartesian components S_x, S_y and S_z.

Same commutation relations that is satisfied by the Cartesian components of the orbital angular momentum.

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z,$$

$$[\hat{S}_y, \hat{S}_z] = i\hbar \hat{S}_x,$$

$$[\hat{S}_z, \hat{S}_x] = i\hbar \hat{S}_y.$$

S^2 commutes with each of the operators S_x, S_y and S_z, that is,

$$[\hat{S}^2, \hat{S}_x] = 0, \quad [\hat{S}^2, \hat{S}_y] = 0, \quad [\hat{S}^2, \hat{S}_z] = 0.$$

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Spin operator S convenient to introduce the *Pauli matrices*

$$\hat{\vec{S}} = \frac{\hbar}{2}\vec{\sigma}.$$

Pauli matrices

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

They have the relations

$$\hat{\sigma}_x\hat{\sigma}_y = -\hat{\sigma}_y\hat{\sigma}_x = i\hat{\sigma}_z,$$

$$\hat{\sigma}_y\hat{\sigma}_z = -\hat{\sigma}_z\hat{\sigma}_y = i\hat{\sigma}_x,$$

$$\hat{\sigma}_z\hat{\sigma}_x = -\hat{\sigma}_x\hat{\sigma}_z = i\hat{\sigma}_y.$$

$$\hat{\sigma}_x\hat{\sigma}_y\hat{\sigma}_z = iI$$

The eigenstate of σ_z is

$$\chi_z^+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_z^- = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

$$\hat{\sigma}_x\chi_z^+ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \chi_z^-,$$

$$\hat{\sigma}_x\chi_z^- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \chi_z^+,$$

$$\hat{\sigma}_y\chi_z^+ = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i\chi_z^-,$$

$$\hat{\sigma}_y\chi_z^- = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -i \\ 0 \end{pmatrix} = -i\chi_z^+$$

Let us consider the addition of the orbital angular momentum and the spin angular momentum, i.e.,

$$\hat{\vec{J}} = \hat{\vec{L}} + \hat{\vec{S}},$$

of a spin half particle (say, of an electron). In the given case l_1 =l(an integer) m_1 =m_l (takes values from –l to l), l_2 =s=1/2, and m_2 =m_s =±1/2.

The value of j in this case is restricted in the interval

$$\left|\ell - \frac{1}{2}\right| \le j \le \left|\ell + \frac{1}{2}\right|.$$

The maximal set of commuting observables in this case is given by:

$$\{\hat{J}^2, \hat{L}^2, \hat{S}^2, \hat{J}_z\}.$$

For the special case of a potential, V (x, y, z), that can be written in the form

$$V(\vec{r}) = V_1(x) + V_2(y) + V_3(z),$$

the three-dimensional TISE reduces to a system of one-dimensional TISE. Indeed, if we write the solution in the form

$$\phi(\vec{r}) = X(x)Y(y)Z(z),$$

therefore,

$$\left[-\frac{\hbar^2}{2m}\frac{1}{X}\frac{d^2X}{dx^2} + V_1(x)\right] + \left[-\frac{\hbar^2}{2m}\frac{1}{Y}\frac{d^2Y}{dy^2} + V_2(y)\right] + \left[-\frac{\hbar^2}{2m}\frac{1}{Z}\frac{d^2Z}{dz^2} + V_3(z)\right] = E,$$

As a result, the solution to the free TISE is given by

$$\psi(x,y,z) = \frac{1}{(2\pi)^{3/2}} e^{ik_1 x} e^{ik_2 y} e^{ik_3 z} = \frac{1}{(2\pi)^{3/2}} e^{i\vec{k}\cdot\vec{r}},$$

The total energy of the particle E is given by the sum of the energy eigenvalues E_1, E_2 and E_3:

$$E = E_1 + E_2 + E_3 = \frac{\hbar^2}{2m}(k_1^2 + k_2^2 + k_3^2) = \frac{\hbar^2}{2m}\vec{k}^2.$$

We note here that the energy, E, depends on the magnitude of the wave vector k but not on its direction. Hence, different orientations of k satisfying the condition

$$|\vec{k}| = \sqrt{k_1^2 + k_2^2 + k_3^2} = \text{const.},$$

Consider a spinless particle confined to move in an infinite rectangular potential well given by

$$V(x,y,z) = \begin{cases} 0 & \text{for } 0 < x < a, 0 < y < b, 0 < z < c \\ +\infty & \text{elsewhere} \end{cases}$$

The normalized solution of the three-dimensional Schrödinger equation

$$\psi_{n_1 n_2 n_3}(x,y,z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_1\pi}{a}x\right) \sin\left(\frac{n_2\pi}{b}y\right) \sin\left(\frac{n_3\pi}{c}z\right), \; n_1, n_2, n_3 = 1, 2, 3, \ldots$$

The corresponding energies are given by

$$E_{n_1 n_2 n_3} = \frac{\hbar^2\pi^2}{2m}\left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2}\right),$$

Consider a particle of mass m moving in a three-dimensional anisotropic oscillator potential

$$\hat{V}(\hat{x},\hat{y},\hat{z}) = \frac{1}{2}m\omega_x^2\hat{X}^2 + \frac{1}{2}m\omega_y^2\hat{Y}^2 + \frac{1}{2}m\omega_z^2\hat{Z}^2.$$

Its Schrödinger equation separates into three equations

$$-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2} + \frac{1}{2}m\omega_x x^2 X(x) = E_x X(x),$$

with similar equations for $Y(y)$ and $Z(z)$. The eigenenergies and wave function corresponding to the potential can be expressed as

$$E_{n_x n_y n_z} = E_{n_x} + E_{n_y} + E_{n_z} = \left(n_x + \frac{1}{2}\right)\hbar\omega_x + \left(n_y + \frac{1}{2}\right)\hbar\omega_y + \left(n_z + \frac{1}{2}\right)\hbar\omega_z,$$

$$\psi_{n_x n_y n_z}(x,y,z) = X_{n_x}(x)Y_{n_y}(y)Z_{n_z}(z),$$

The radial equation for a given l for R_l can be written as

$$\frac{d}{dr}\left(r^2\frac{dR_{n\ell}}{dr}\right)-\frac{2mr^2}{\hbar^2}(V(r)-E)R_{n\ell}=l(l+1)R_{n\ell},$$

This equation can be simplified further by changing the variables:

$$u_{n\ell}(r)=rR_{n\ell}(r).$$

Therefore,

$$\frac{d^2u_{n\ell}}{dr^2}+\frac{2m}{\hbar^2}\left[E-V(r)-\frac{\hbar^2\ell(\ell+1)}{2mr^2}\right]u_{n\ell}(r)=0.$$

This equation for the radial function $u_{nl}(r)$ can be solved only if the potential, V (r), is prescribed.

As a result, we have

$$\phi_{k\ell m}(\rho,\theta,\varphi) = A_\ell j_\ell(kr) Y_\ell^m(\theta,\varphi),$$

Recall that the free particle solution in Cartesian coordinates is proportional to plane wave, which can be expanded in terms of the spherical Bessel functions as

$$e^{i\vec{k}\cdot\vec{r}} = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} c_{\ell m}\, j_\ell(kr)\, Y_\ell^m(\theta,\varphi),$$

For the particular case of propagation along the z-axis (k∥z), m = 0 and we get

$$e^{i\vec{k}\cdot\vec{r}} = e^{kr\cos\theta} = \sum_{\ell=0}^{\infty} i^\ell (2\ell+1) j_\ell(kr) P_\ell(\cos\theta), \qquad c_{\ell m} = i^\ell (2\ell+1).$$

The stationary state energies and the corresponding wave functions of the hydrogen atom are, respectively,

$$E_n = -\frac{\mu}{2\hbar^2}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \frac{1}{n^2}, (n = 1,2,3,...),$$

$$\psi_{n\ell m}(r,\theta,\varphi) = \phi_{n\ell m}(r,\theta,\varphi)e^{-\frac{i}{\hbar}E_n t} = R_{n\ell}(r)Y_\ell^m(\theta,\varphi)e^{-\frac{i}{\hbar}E_n t},$$

$$R_{n\ell}(r) = e^{-r/na_0}\left(\rho^\ell \sum_{j=0}^{n_r} c_j \rho^j\right) = e^{-r/na_0}\left(\frac{2r}{na_0}\right)^\ell \sum_{j=0}^{n_r} c_j \left(\frac{2r}{na_0}\right)^j,$$

$$c_{j+1} = \frac{[(\ell+j+1)-n]}{(\ell+j+1)(\ell+j+2)-\ell(\ell+1)}c_j,$$

The degeneracy g is given by

$$g = \sum_{\ell=0}^{n-1}(2\ell+1) = \frac{n}{2}[2\times 1 + (n-1)\times 2] = n^2.$$

The energy eigenvalues of the oscillator associated with a given value of l:

$$E_{k,\ell} = \hbar\omega\left(k+\ell+\frac{3}{2}\right),$$

The degeneracy of the energy levels is calculated to be

$$g_{n_{even}} = \sum_{\ell=0,2,4,\ldots,n}(2\ell+1) = \frac{(n+1)(n+2)}{2}, \qquad \text{for even } n,$$

$$g_{n_{odd}} = \sum_{\ell=1,3,5,\ldots,n}(2\ell+1) = \frac{(n+1)(n+2)}{2}, \qquad \text{for odd } n.$$

Ground state: The ground state corresponds to n = 0 for which l = k = 0.

$$\phi_{000} = \frac{2}{\pi^{1/4}}\left(\frac{m\omega}{\hbar}\right)^{3/4} e^{-\frac{m\omega}{2\hbar}r^2} Y_{00}(\theta,\varphi).$$

The separation of variables leading to N independent single-particle Schrödinger equations

$$-\frac{\hbar^2}{2m_j}\vec{\nabla}_j^2\phi(\vec{r}_j)+V(\vec{r}_j)\phi_j(\vec{r}_j)=E_j\phi_j(\vec{r}_j), \qquad j=1,2,3,...,N.$$

The stationary state wave function of the system is then given by the product of the single-particle wave functions

$$\psi(\vec{r}_1,...,\vec{r}_N,t)=\phi_1(\vec{r}_1)\phi_2(\vec{r}_2)\phi_3(\vec{r}_3)...\phi_N(\vec{r}_N)\ e^{-\frac{i}{\hbar}(E_1+E_2+E_3+...+E_N)t}$$

$$=\left(\prod_{j=1}^{N}\phi_j(\vec{r}_j)\right)\ e^{-\frac{i}{\hbar}Et},$$

with energy

$$E=E_1+E_2+E_3+...+E_N=\sum_{j=1}^{N}E_j.$$

Since the particles are indistinguishable, no experiment can determine which of the particles of the system is at r_j and which one is at r_k. The probability density, therefore, should remain unchanged, that is,

$$\phi(\vec{r}_1,\vec{r}_2,...,\vec{r}_j,...,\vec{r}_k,...,\vec{r}_N) = \pm\phi(\vec{r}_1,\vec{r}_2,...,\vec{r}_k,...,\vec{r}_j,...,\vec{r}_N).$$

The symmetric and the anti-symmetric wave functions, $\phi_s(\xi_1, \xi_2, \xi_3,..., \xi_N)$ and $\phi_a(\xi_1, \xi_2, \xi_3, ..., \xi_N)$ respectively, of the system of N identical and indistinguishable particles are written as

$$\phi_s(\xi_1,\xi_2,\xi_3,...,\xi_N) = \frac{1}{\sqrt{N!}}\sum_P P\left\{\phi_{n_1}(\xi_1)\phi_{n_2}(\xi_2)\phi_{n_3}(\xi_3)...\phi_N(\xi_N)\right\},$$

$$\phi_a(\xi_1,\xi_2,\xi_3,...,\xi_N) = \frac{1}{\sqrt{N!}}\sum_P (-1)^P P\left\{\phi_{n_1}(\xi_1)\phi_{n_2}(\xi_2)\phi_{n_3}(\xi_3)...\phi_N(\xi_N)\right\}$$

In general, the N-particle anti-symmetric wave function can be written as

$$\phi_a(\xi_1,\xi_2,\xi_3,...,\xi_N)=\frac{1}{\sqrt{N!}}\begin{vmatrix}\phi_{n_1}(\xi_1) & \phi_{n_1}(\xi_2) & \cdots & \phi_{n_1}(\xi_N)\\ \phi_{n_2}(\xi_1) & \phi_{n_2}(\xi_2) & \cdots & \phi_{n_2}(\xi_N)\\ \cdots & \cdots & \cdots & \cdots\\ \cdots & \cdots & \cdots & \cdots\\ \phi_{n_N}(\xi_1) & \phi_{n_N}(\xi_2) & \cdots & \phi_{n_N}(\xi_N)\end{vmatrix}.$$

This is known as the Slater determinant. Note that interchanging any two identical particles is equivalent to interchanging the corresponding columns of the Slater determinant. From the properties of the determinants, we know that interchanging two columns of a determinant results in the multiplication of the determinant by (–1).

For a system of N identical fermions, the wave function must be overall anti-symmetric

$$\phi_a(\vec{r}_1,...,\vec{r}_N,\vec{S}_1,...,\vec{S}_1) = \begin{cases} \phi_s(\vec{r}_1,...,\vec{r}_N)\,\chi_a(\vec{S}_1,...,\vec{S}_N) \\ \phi_a(\vec{r}_1,...,\vec{r}_N)\,\chi_s(\vec{S}_1,...,\vec{S}_N). \end{cases}$$

For two particles

$$\chi_a = \frac{1}{\sqrt{2}}\left\{\uparrow_1\downarrow_2 - \uparrow_2\downarrow_1\right\} m_s = 0;\ S = 0,\ \text{singlet state}$$

$$\chi_s = \left[\begin{array}{cc} \uparrow_1\uparrow_2 & m_s = 1 \\ \frac{1}{\sqrt{2}}\left\{\uparrow_1\downarrow_2 + \uparrow_2\downarrow_1\right\} & m_s = 0 \\ \downarrow_1\downarrow_2 & m_s = -1 \end{array}\right];\ S = 1,\ \text{triplet state}$$

The Hamiltonian for the one-dimensional harmonic oscillator (a particle of mass m attached to a spring) is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2\hat{x}^2,$$

Let us introduce the following operators

$$\hat{a} = \frac{1}{\sqrt{2m\hbar\omega}}(i\hat{p} + m\omega\hat{x}), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2m\hbar\omega}}(-i\hat{p} + m\omega\hat{x}).$$

We get that

$$\hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$$

The following commutation relation between a and a+

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

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The generalized equation of time-independent perturbation theory is

$$H_0\left|\psi_n^{(k)}\right\rangle + H_1\left|\psi_n^{(k-1)}\right\rangle = \sum_{i=0}^{k} E_n^{(i)}\left|\psi_n^{(k-i)}\right\rangle$$

The first order energy and wave function is

$$E_n^{(1)} = \left\langle \psi_n^{(0)} \middle| H_1 \middle| \psi_n^{(0)} \right\rangle \qquad \left|\psi_n^{(1)}\right\rangle = \sum_{m\neq n} \frac{\left\langle \psi_m^{(0)} \middle| H_1 \middle| \psi_n^{(0)} \right\rangle}{\left(E_m^{(0)} - E_n^{(0)}\right)} \left|\psi_m^{(0)}\right\rangle$$

The second order energy and wave function is

$$E_n^{(2)} = \sum_{m\neq n} \frac{\left\langle \psi_m^{(0)} \middle| H_1 \middle| \psi_n^{(0)} \right\rangle \left\langle \psi_n^{(0)} \middle| H_1 \middle| \psi_m^{(0)} \right\rangle}{\left(E_m^{(0)} - E_n^{(0)}\right)}$$

$$\left|\psi_n^{(2)}\right\rangle = \sum_{\nu\neq n}\sum_{\mu\neq n}\left[\frac{\left\langle \psi_\mu^{(0)} \middle| H_1 \middle| \psi_n^{(0)} \right\rangle \left\langle \psi_\nu^{(0)} \middle| H_1 \middle| \psi_\mu^{(0)} \right\rangle}{\left(E_\mu^{(0)} - E_n^{(0)}\right)\left(E_n^{(0)} - E_\nu^{(0)}\right)} - \frac{\left\langle \psi_n^{(0)} \middle| H_1 \middle| \psi_n^{(0)} \right\rangle \left\langle \psi_\mu^{(0)} \middle| H_1 \middle| \psi_n^{(0)} \right\rangle}{\left(E_\mu^{(0-)} - E_n^{(0)}\right)^2}\right]\left|\psi_\mu^{(0)}\right\rangle$$

$$-\frac{1}{2}\sum_{\nu\neq n}\frac{\left|\left\langle \psi_\nu^{(0)} \middle| H_1 \middle| \psi_n^{(0)} \right\rangle\right|^2}{\left(E_\nu^{(0)} - E_n^{(0)}\right)^2}\left|\psi_n^{(0)}\right\rangle$$

On defining

$$\left\langle \psi_{n\beta}^{(0)} \middle| H_1 \middle| \psi_{n\alpha}^{(0)} \right\rangle = h_{\beta\alpha}$$

and then varying both α and β between 1 and r, we have

$$\begin{bmatrix} h_{11}-E_n^{(1)} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22}-E_n^{(1)} & \cdots & h_{2r} \\ \vdots & \vdots & \vdots & \vdots \\ h_{r1} & h_{r2} & \cdots & h_{rr}-E_n^{(1)} \end{bmatrix} \begin{bmatrix} D_1 \\ D_2 \\ \vdots \\ D_r \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Its non-trivial solution demands:

$$\begin{vmatrix} h_{11}-E_n^{(1)} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22}-E_n^{(1)} & \cdots & h_{2r} \\ \vdots & \vdots & \vdots & \vdots \\ h_{r1} & h_{r2} & \cdots & h_{rr}-E_n^{(1)} \end{vmatrix} = 0$$

In solving the problems where it is hard to determine a good unperturbed Hamiltonian, to make the perturbation small and solvable, the variational method is more robust in comparison with the perturbation theory.

The variational theorem states that the expectation value of the Hamiltonian <H> calculated with the use of a normalized trial wave function ket $|\phi\rangle$ is always greater than or equal to E_g:

$$E_g \le \langle H \rangle = \langle \phi | H | \phi \rangle$$

where

$$\langle \phi | \phi \rangle = 1.$$

We then have,

$$\langle \psi_a | H | \psi_a \rangle = \sum_{i=1}^{N} \Big[\int \phi_{n_i}^*(\mathbf{r}_i) \Big\{ -\frac{\hbar^2}{2m}\nabla_i^2 + V_{ext}(\mathbf{r}_i) \Big\} \phi_{n_i}(\mathbf{r}_i) d^3 r_i$$

$$+ \frac{1}{2}\sum_{i\neq j} \iint |\phi_{n_i}(\mathbf{r}_i)|^2 |\phi_{n_j}(\mathbf{r}_j)|^2 \frac{e^2}{4\pi\varepsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} d^3 r_i d^3 r_j$$

$$- \frac{1}{2}\sum_{i\neq j} \iint \phi_{n_i}^*(\mathbf{r}_i)\phi_{n_j}^*(\mathbf{r}_j) \frac{e^2}{4\pi\varepsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} \phi_{n_j}(\mathbf{r}_i)\phi_{n_i}(\mathbf{r}_j) d^3 r_i d^3 r_j \Big]$$

Minimization with respect to $\phi^*_{n_i}(\mathbf{r}_i)$,

$$\Big\{ -\frac{\hbar^2}{2m}\nabla_i^2 + V_{ext}(\mathbf{r}_i) \Big\} \phi_{n_i}(\mathbf{r}_i) + \frac{e^2}{4\pi\varepsilon_0} \Big\{ \sum_{j\neq i} \int |\phi_{n_j}(\mathbf{r}_j)|^2 \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} d^3 r_j \Big\} \phi_{n_i}(\mathbf{r}_i)$$

$$- \frac{e^2}{4\pi\varepsilon_0} \Big\{ \sum_{j\neq i} \int \phi_{n_j}^*(\mathbf{r}_j) \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \phi_{n_i}(\mathbf{r}_j) d^3 r_j \phi_{n_j}(\mathbf{r}_i) \Big\} = \varepsilon_{n_i} \phi_{n_i}(\mathbf{r}_i)$$

This equation is known as the Hartree-Fock equation.

The third term on the left is originated from exchange interactions.

$$i\hbar\sum_n \dot{c}_n(t)e^{\frac{i(E_m-E_n)t}{\hbar}}\langle\phi_m|\phi_n\rangle = \sum_n c_n(t)e^{\frac{i(E_m-E_n)t}{\hbar}}\langle\phi_m|H_1(\mathbf{r},t)|\phi_n\rangle$$

$$\Rightarrow \dot{c}_m(t) = \frac{1}{i\hbar}\sum_n h_{mn}(t)c_n(t)e^{i\omega_{mn}t}$$

We thus obtained a matrix equation,

$$\frac{d}{dt}\begin{bmatrix} c_1 \\ c_2 \\ . \\ . \end{bmatrix} = \frac{1}{i\hbar}\begin{bmatrix} h_{11}e^{i\omega_{11}t} & h_{12}e^{i\omega_{12}t} & . & . \\ h_{21}e^{i\omega_{21}t} & h_{22}e^{i\omega_{22}t} & . & . \\ . & . & . & . \\ . & . & . & . \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \\ . \\ . \end{bmatrix}$$

which yields coupled equations that are solved to find $c_1(t)$, $c_2(t)$.....$c_n(t)$. After knowing $c_1(t)$, $c_2(t)$.....$c_n(t)$..... , one determines the probability of finding the system in any particular state at a later time.

To satisfy this, the coefficient of each power of λ must vanish. We therefore have,

$$\frac{dc_m^{(0)}(t)}{dt} = 0$$

$$\frac{dc_m^{(1)}(t)}{dt} = \frac{1}{i\hbar}\sum_n h_{mn}(t)c_n^{(0)}(t)e^{i\omega_{mn}t}$$

$$\frac{dc_m^{(2)}(t)}{dt} = \frac{1}{i\hbar}\sum_n h_{mn}(t)c_n^{(1)}(t)e^{i\omega_{mn}t}$$

The first order contribution to $c_m(t)$ is

$$c_m^{(1)}(t) = \frac{1}{i\hbar}\int\sum_n h_{mn}(t')c_n^{(0)}(t')e^{i\omega_{mn}t'}\,dt'$$

If the system initially at t =0 is in only one unperturbed state ϕ_n then $c_n^{(0)}$ is non-zero (equal to 1) only for one value of n belonging to ϕ_n, and it is zero for all other values of m.

The number of particles scattered into an element of solid angle

$$d\Omega = \sin\theta \, d\theta \, d\varphi$$

is proportional to a quantity that plays a central role in the physics of scattering: the *differential cross section*. The differential cross section, which is is defined as the number of particles scattered into an element of solid angle $d\Omega$ in the direction (θ,φ) per unit time and incident flux:

$$\frac{d\sigma(\theta,\varphi)}{d\Omega} = \frac{1}{J_{inc}} \frac{dN(\theta,\varphi)}{d\Omega},$$

where J_{inc} is the incident flux (or incident current density); it is equal to the number of incident particles per area per unit time.

In general, however, the scattered wave is not spherically symmetric; its amplitude depends on the direction (θ,φ) along which is detected and hence

$$\phi_{sc}(\vec{r}) = Af(\theta,\varphi)\frac{e^{i\vec{k}\cdot\vec{r}}}{r},$$

Therefore, the differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{1}{J_{inc}}\frac{dN}{d\Omega} = \frac{k}{k_0}\left| f(\theta,\varphi)\right|^2.$$

Since the normalization factor A does not contribute to the differential cross section, we will be taking it equal to one. For elastic scattering k_0 is equal to k

$$\frac{d\sigma}{d\Omega} = \left| f(\theta,\varphi)\right|^2.$$

We obtain the total scattered wave function:

$$\psi(\vec{r}) = \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|} V(\vec{r}\,')\psi(\vec{r}\,')\, d^3r'.$$

If the potential $V(r)$ is weak enough, it will distort only slightly the incident plane wave. The *first Born approximation* consists then of approximating the scattered wave function by a *plane wave*.

$$\psi(\vec{r}) \simeq \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|} V(\vec{r}\,')\phi_{inc}(\vec{r}\,')\, d^3r'.$$

Thus, we can write the scattering amplitude and the differential cross section in the first Bonn approximation as

$$\frac{d\sigma}{d\Omega} = \left| f(\theta,\varphi)\right|^2 = \frac{\mu^2}{4\pi^2\hbar^4}\left|\int e^{i\vec{q}\cdot\vec{r}\,'} V(\vec{r}\,')\, d^3r'\right|^2.$$

First, since the limit of the Bessel function $j_l(kr)$ for large values of r is given by

$$j_l(kr) \longrightarrow \frac{\sin(kr - l\pi/2)}{kr} \qquad (r \longrightarrow \infty),$$

The asymptotic form of wave function becomes

$$\psi(r,\theta) \longrightarrow \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos\theta) \frac{\sin(kr - l\pi/2)}{kr} + f(\theta)\frac{e^{ikr}}{r},$$

Since

$$\sin(kr - l\pi/2) = [(-i)^l e^{ikr} - i^l e^{-ikr}]/2i$$

Therefore

$$\psi(r,\theta) \longrightarrow -\frac{e^{-ikr}}{2ikr}\sum_{l=0}^{\infty} i^{2l}(2l+1)P_l(\cos\theta) + \frac{e^{ikr}}{r}\left[f(\theta) + \frac{1}{2ik}\sum_{l=0}^{\infty} i^l(-i)^l(2l+1)P_l(\cos\theta)\right]$$